ECON 0100 | HW C1 | Externalities

Due: Friday, Oct. 16

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. You are welcomed and encouraged to work in groups as long as your work is your own. Submit your answers on Gradescope when you're finished: each answer there is a selection.

Q1 | Malodorous Byproducts

Butterbeer is beloved by Hogwarts students and teachers alike and is widely available at wizarding shops. But the beverage has the negative effect of generating displeasing odors from its imbibers. Preferences for butterbeer and its marginal cost can be represented by the following relationships.

$$D : P_B = 100 - \frac{1}{2}Q_D \qquad S : P_S = 20 + \frac{1}{2}Q_S$$

Prices are in galleons and quantity is in bottles. The odor imposes a cost of 20 galleons per bottle on everyone nearby.

Marginal Social Cost: _____

Equilibrium Price ($\hat{P}$): _____ Equilibrium Quantity ($\hat{Q}$): _____

Efficient Quantity (Q^*): _____ Deadweight Loss (DWL): _____

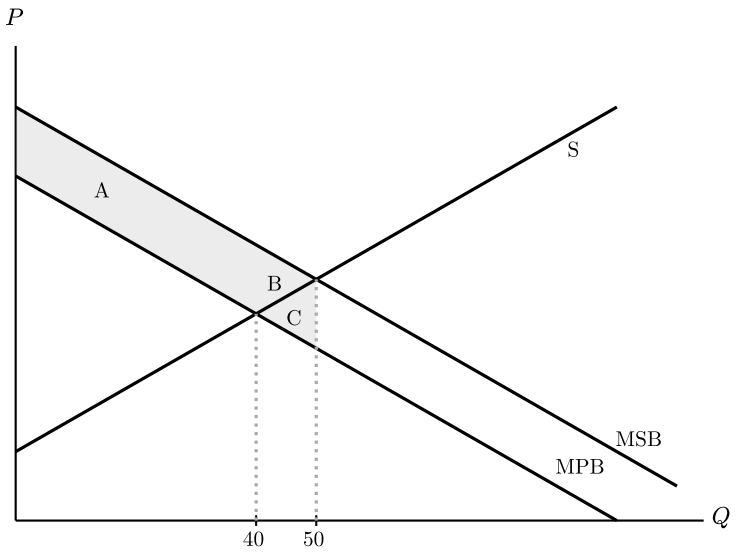
Q2 | Covid Vaccine

Lets model Covid and assume the following.

1. There is a linear marginal private benefit curve for the vaccine.
2. The vaccine carries with it a positive externality.
3. There is a linear upward sloping supply curve for the vaccine. The supply curve for the vaccine is a little funny, and we'll talk about imperfect close substitutes in the next few weeks. But we'll just assume it's there.

Use a graph to discuss the welfare in the market and any potential market failure.

The graph below shows the market. The upper downward-sloping curve is marginal social benefit and the lower one is marginal private benefit. Supply is both the private and the social cost. The lettered regions are for the questions.



Market Quantity ($\hat{Q}$): _____ Efficient Quantity (Q^*): _____

Deadweight Loss (letter): _____

Why does the market provide too few vaccinations?

a) Each vaccination costs society more to produce than it costs the seller.

b) Each vaccination is worth more to society than to the person buying it, so at the market quantity marginal social benefit is still above marginal cost.

c) Buyers value each vaccination less than sellers' marginal cost, so too few buyers enter the market.

d) The market quantity is already socially efficient, so no vaccinations are missing.